Educational Psychology Review (2022) 34:989–1024
https://doi.org/10.1007/s10648-021-09654-1

META-ANALYSIS

Effects of Learner-Generated Highlighting and Instructor-Provided Highlighting on Learning from Text: A Meta-Analysis

Héctor R. Ponce[1] · Richard E. Mayer[2] · Ester E. Méndez[3]

Accepted: 20 December 2021 / Published online: 7 January 2022

Abstract
The present study examines the existing published research about the effectiveness of learner-generated highlighting and instructor-provided highlighting on learning from text. A meta-analysis was conducted of scientifically rigorous experiments comparing the learning outcomes (i.e., performance on memory and/or comprehension tests) of students (i.e., college students and/or K-12 students) who read an academic text with or without being asked to highlight important material (i.e., with or without learner-generated highlighting) or who read an academic text with or without the important material already being highlighted (i.e., with or without instructor-provided highlighting). We found 36 published articles that met these criteria ranging from the years 1938 to 2019, which generated 85 effect sizes. The results showed that learner-generated highlighting improved memory but not comprehension, with average effect sizes of 0.36 and 0.20, respectively; and instructor-provided highlighting improved both memory and comprehension, both with an average effect size of 0.44. Learner-generated highlighting improved learning for college students but not for school students, with average effect sizes of 0.39 and 0.24, respectively; and instructor-provided highlighting improved learning for both college and school students, with average effect sizes of 0.41 and 0.48, respectively. We discuss the theoretical and practical implications of these findings.

Keywords Meta-analysis · Highlighting · Learner-generated highlighting · Instructor-provided highlighting · Learning strategy

✉ Héctor R. Ponce
hector.ponce@usach.cl

Extended author information available on the last page of the article

The Steamboat

Eastern-style steamboats became a financial success in 1807. These one-story boats operated on the Hudson River and other eastern rivers. These rivers were deep and suited perfectly the deep hulls of the eastern steamboat. The cargo was stored in these deep hulls below the main deck. The eastern steamboats used low-pressure engines. Western-style steamboats, however, were different. They churned their way up the shallow waters of the Missouri, Ohio, and Mississippi Rivers. Their hulls were flat, without room for cargo. The cargo was carried on the main deck or on the superstructure, one or two floors above the main deck. More efficient and dangerous high-pressure engines were used and often burned up to 32 cords of wood a day.

Fig. 1 The Steamboat Passage

Objective and Rationale

Suppose a student is reading an academic text on steamboats such as shown in Fig. 1, taken from a study by Ponce and Mayer (2014a). How can we help guide the learners' active cognitive processing during learning in an effort to improve their learning outcome? One approach is to ask the learner to engage in a learning strategy—that is, an activity that learners engage in during learning with the intention of improving their learning outcome. For example, in this learning strategy approach, we could ask the learner to highlight the important material in the lesson. With a paper-based medium, highlighting can be accomplished by asking the learner to use a colored marker to add color to the important text or to use a pencil to underline important text; with a computer-based medium, highlighting can be accomplished by asking the learner to use a highlighting tool that adds background color to the important text, or underlines important text, or puts important text in bold or italicized or colored font. The act of highlighting is intended to activate the learners' cognitive process of selecting relevant information for processing in working memory on its way to storage in long-term memory (Fiorella & Mayer. 2015). We call this approach learner-generated highlighting.

Another approach is to design instructional text so that important material is already highlighted. For example, in this instructional design approach, the passage is presented to the learner with important parts already highlighted based on the instructional goal of the lesson. With a paper-based or computer-based medium, the important text can be highlighted by adding background color, adding underlining, or using bold or italicized or colored font. Figure 2 provides an example of the steamboat text with important elements in bold font, based on the judgment of the instructional designer. Providing highlighting is intended to efficiently guide the learner's cognitive process of selecting relevant information for processing in working memory, thereby freeing up cognitive capacity that can be used to mentally organize this important information and integrate it with relevant prior knowledge (Fiorella & Mayer. 2015, 2016). We call this approach instructor-provided highlighting.

Instructor-provided highlighting can be considered a component of signaling, which involves adding cues that guide the learner's attention to essential material in a lesson or to the organization of the essential material (Mayer. 2021; van Cog. 2022). In particular, instructor-provided highlighting is a type of text-based cue (van Cog. 2022) or verbal signaling (Mayer. 2021) because it involves guiding attention to certain words. Other forms of verbal signaling that can be used in conjunction with highlighting include providing an outline, headings, pointer words (such as, "first…second…third"), and graphic organizers (Mayer. 2021; Ponce et al. 2018a).

Thus, the goal of the present study is to examine what existing published research has to say about the effectiveness of learner-generated highlighting and instructor-provided highlighting on learning from text. In particular, we conduct a meta-analysis of scientifically rigorous experiments comparing the learning outcomes (i.e., performance on memory and/or comprehension tests) of students (i.e., college students and/or K-12 students) who read an academic text with or without being asked to highlight important material (i.e., with or without learner-generated highlighting) or who read an academic text with or without the important material already being highlighted (i.e., with or without instructor-provided highlighting).

Our rationale for choosing to focus on highlighting is that it is a commonly used and easy to use learning strategy (Miyatsu et al. 2018) or instructional design feature (Mayer. 2021). However, in contrast, highlighting is considered one of the least effective and least valued learning strategies (Dunlosky et al. 2013; Fiorella & Mayer. 2015, 2016) or instructional design features (Mayer. 2021; Ponce & Mayer. 2014a). For instance, Dunlosky et al.'s (2013) review is particularly critical of the usefulness of the highlighting strategy, as they indicate: "most studies

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Fig. 2 The Steamboat Passage Highlighted

have shown no benefit of highlighting (as it is typically used) over and above the benefit of simply reading, and thus the question concerning the generality of the benefits of highlighting is largely moot" (p. 20). However, in a more recent review, Miyatsu et al. (2018) argue that although the evidence seems to indicate that highlighting and rereading may not be effective in educational contexts, such argument overlooks the possibility that these strategies may be effective in specific circumstances. The reason is that another way to scrutinize these strategies is to compare their effectiveness relative to no study strategies, such as when students only read the text.

Overall, we chose to focus on highlighting because we wanted to closely examine whether the current state of the literature could shed light on the conditions—if any—under which this lowliest of learning strategies and most underappreciated of instructional design features could lead to better learning outcomes. In examining this underdog of learning aids, we wish to determine whether there are ways of using highlighting to benefit student learning.

Theoretical Framework and Hypotheses

We ground our search in generative learning theory (Fiorella & Mayer. 2015; Wittrock. 1989) and the cognitive theory of multimedia learning (Mayer. 2021), which posit that active learning occurs when learners engage in appropriate cognitive processing during learning. As shown in the left columns of Table 1 and Table 2, three cognitive processes required for meaningful learning from text are selecting (i.e., attending to the relevant information in a text and transferring that information to working memory), organizing (i.e., mentally organizing the incoming information into a coherent cognitive representation in working memory), and integrating (i.e., connecting incoming information into working memory with relevant prior knowledge activated from long-term memory). When the learner engages in none of these processes, the learning outcome is no learning, indicated by poor performance on tests of memory and comprehension. When the learner engages mainly in selecting, the learning outcome is rote learning, indicated by good performance on tests of memory but not tests of comprehension. When the learner engages in all three cognitive processes during learning, the learning outcome is meaningful learning, reflected in good performance on both memory and comprehension tests.

Consider what happens when the learner does or does not engage in learner-generated highlighting, as summarized in the top of Table 1. The middle column of Table 1 shows how learner-generated highlighting relates to these three processes. First, learner-generated highlighting activates the cognitive process of selecting because the learner must search for the important information in a text that needs to be highlighted, but processing capacity is limited, so when the learner uses too much processing capacity on the cognitive process of selecting there may insufficient capacity available for organizing and integrating. Thus, as shown in the right column of Table 1, we predict that student-generated highlighting will yield improvements in tests of memory (which reflect the fruits of engaging in selecting) but not in tests of comprehension (which reflect the fruits of engaging in all three cognitive processes; Hypothesis 1). Second, if the learner is inexperienced in highlighting, the learner may select unimportant material information and will attempt to engage in organizing and integrating with that unimportant material. Thus as shown in the right side of Table 1, we predict that student-generated highlighting will yield improvements for college students (who presumably have developed some proficiency in highlighting important information) but not for K-12 students (who presumably have no developed adequate proficiency in highlighting; Hypothesis 2). Third, if the learner is inefficient in highlighting, the learner will select unimportant information and will attempt to engage in organizing and integrating with that

Table 1 Cognitive Processes and Predictions for Learner-Generated Highlighting

Cognitive Processes	Explanations	Hypotheses
Selecting	Highlighting focuses cognitive capacity on the process of selecting, leaving less capacity for the cognitive processes of organizing and integrating	1. Learner-generated highlighting works for memory tests but not for comprehension tests
Selecting	Less experienced learners may not know what to highlight and select unimportant information affecting organizing and integrating processes	2. Learner-generated highlighting works for college-students but not K-12 students
Selecting, organizing, and integrating	Less experienced learners may be inefficient in highlight and select unimportant information affecting organizing and integrating processes	3. Learner-generated highlighting is enhanced by training
Selecting, organizing, and integrating	Highlighting is one step in activating appropriate cognitive processing during learning	4. Learner-generated highlighting is enhanced by adding a complementary learning strategy
None	Reviewing of inappropriate highlighting is not helpful	5. Learner-generated highlighting is not enhanced by reviewing

unimportant material. Thus, as shown in the right side of Table 1, we predict that learner-generated highlighting will yield improvements in learning outcomes when students are given training in how to highlight but not when they receive no training (Hypothesis 3). Fourth, given the potential of errors in selecting relevant material and the possibility that students might not be able to follow up with deeper processing of organizing and integrating, we predict student-generated highlighting will yield improvements in learning outcomes when a complementary learning strategy is added (such as notetaking or constructing a graphic organizer), as shown in the right side of Table 1 (Hypothesis 4). Fifth, if students select the wrong information to highlight, then we predict reviewing what they have highlighted will not improve learning outcome performance as shown in the right side of Table 1 (Hypothesis 5).

Next, consider what happens when the learner does or does not receive instructor-provided highlighting, as summarized in Table 2. The middle column of Table 2 shows how instructor-provided highlighting relates to the three cognitive processes and the rightmost column summarizes our predictions concerning learning outcomes.

Table 2 Cognitive Processes and Predictions for Instructor-Provided Highlighting

Cognitive Processes	Explanations	Hypotheses
Selecting, organizing, and integrating	High-quality highlighting guides the cognitive process of selecting and frees up cognitive capacity for the cognitive processes of organizing and integrating	6. Instructor-provided highlighting works for memory and comprehension tests
Selecting, organizing, and integrating	High-quality highlighting overcomes the problem that inexperienced students are likely select the wrong material	7. Instructor-provided highlighting works for college students and K-12 students

First, instructor-provided highlighting guides the learner's selecting processes with high efficiency, so there is also cognitive capacity available for the higher-level processes of organizing and integrating. Thus, as shown in the right side of Table 2, we predict instructor-provided highlighting will lead to improvements in both memory (reflecting the role of selecting) and comprehension tests (reflecting the impact of all three cognitive processes; Hypothesis 6). Second, instructor-provided highlighting overcomes the problem that inexperienced students are likely to select the wrong material, so we predict that instructor-provided highlighting will be helpful for both college students and K-12 students, as shown in the right column of Table 2 (Hypothesis 7).

It could be argued that instructor-provided highlighting inhibits the learner's selecting process, because the learner does not have to decide what is important. In other words, it could be argued that when selecting is done for the learner, this deprives the learner of the benefits of generative learning. This analysis makes sense if we assume that learners have sufficient cognitive capacity to allow them to engage in selecting, organizing, and integrating processes during learning. In contrast, when learners must devote a large portion of their limited processing capacity to deciding what is important in a lesson (i.e., selecting), we assume that they may not have sufficient capacity remaining for engaging in the processes that are central to deep learning—mentally organizing that information and relating it to relevant prior knowledge. Additionally, some learners may select inappropriate information, which eliminates the benefits of engaging in organizing and integrating processes. Thus, our general prediction is that learner-generated highlighting will improve performance both on memory tests (which reflect the learner engaging in the selecting process during learning) and comprehension tests (which reflect the learner engaging in organizing and integrating processes during learning).

Method

We searched and examined published research on learner-generated highlighting and instructor-provided highlighting, focusing on studies that reported experiments with a control group, measured learning outcomes, such as performance on memory and/or comprehension tests, and involved college students or K-12 students as participants.

Inclusion and Exclusion Criteria

Seven criteria were considered for the selection of studies:

1. The study focused on the highlighting strategy, including any of the related typographical cues used to mark text such as underlining, yellow highlighting, capitalizing, among others.

2. We also included studies that examined highlighting in combination with another strategy, such as asking students to highlight text and fill in a graphic organizer. To select this type of studies, we examined each experimental design and analyzed whether highlighting was used in conjunction with another strategy; we found several studies of this type (see Table 4).
3. The treatment group was instructed to highlight part of the text (i.e., learner-generated highlighting) or the treatment group was asked to study a text already highlighted by the experimenter (i.e., instructor-provided highlighting).
4. The control group was asked to read or study an academic text only. Studies were not included in the analysis if the control group was given an alternative learning activity.
5. The study included at least one posttest that measured memory or reading comprehension.
6. Studies were not included if their statistical results were not sufficient to compute a weighted effect size.
7. We included studies in English and published in journals or conference proceedings.

Literature Search Strategies

The literature search started with a snowball approach by recollecting the studies referred to by Miyatsu et al. (2018) and Dunlosky et al. (2013) in those sections dedicated to reviewing the highlighting strategy. Additionally, we examined the meta-analysis on signaling with media conducted by Schneider et al. (2018), which included an analysis of organizational highlighting as a mode of text signaling. Miyatsu et al. (2018) and Dunlosky et al. (2013) cited 40 and 39 studies, respectively, and Schneider et al. (2018) cited 36 studies. After removing 29 duplicate references, a total of 86 unique studies were selected for detailed screening.

In a second step, we searched for studies in the Web of Science database. In the "advanced search" option, different combinations of words were employed in the TI (title), TS (topic), and AB (abstract) options. The words used were "text marking," "highlighting AND comprehension," "underlining AND comprehension," "highlighting AND reading," "underlining AND reading," "highlighting AND strategy," "underlining AND strategy," "text marking AND study strategy," and "typographical cues." The results were refined by searching within the following databases: multidisciplinary sciences, educational research, arts, computer science information systems, humanities multidisciplinary, psychology multidisciplinary, social sciences interdisciplinary, educational sciences disciplines, language linguistics, literary, history social science. These searches generated 1,523 records. After reviewing the titles of each study, we concluded that 1,471 records were not related to the highlighting strategy. Thus, 52 studies were analyzed; we found 32 duplicate studies, adding 20 new studies to the previous 86 references, totalizing thus 106 studies. It is essential to mention that a more general search with broader concepts such as "highlighting," "underlining," "highlighting OR underlining" resulted in more than two hundred thousand titles with many articles from areas such as biology and chemistry.

Additionally, 39 studies were found by using other sources such as ProQuest and suggestions provided by Mendeley reference management software and by searching the references cited in more recent studies on highlighting (Ben-Yehudah & Eshet-Alkalai. 2018; Ponce et al., 2018b; Ponce et al. 2018a; Winchell et al. 2020). Two of these studies were duplicates of the previous selections.

The literature search included all studies available through March 2021 and generated a total of 143 unique studies selected for detailed screening. From this list, eight studies were not considered in detail for the following reasons: two were not found, two were in German, one was a book chapter, and three were books. We classified the remaining 135 studies according to whether they were experimental or not, discarding 19 studies. Additionally, 43 studies were not considered since they focused on other learning or study strategies. The remaining 73 articles were thoroughly reviewed by isolating the data needed to compute effect sizes. We computed effect sizes for memory and comprehension tests. Studies that included only other types of tests, such as reading speed, vocabulary, and frequency of text marking, were excluded from the meta-analysis, resulting in 10 studies being removed.

Fifteen studies with insufficient data to compute effect sizes and four studies without a control group (i.e., read-only group) were also excluded. Finally, we excluded three more studies that tested comprehension from diagrams and not from texts, and other five studies that were postgraduate dissertations rather than published articles. Consequently, 36 studies were considered in the meta-analysis. Figure 3 presents a summary of the literature search strategies based on the PRISMA flowchart (Moher et al. 2009). From these studies, 28 articles examined learner-generated highlighting and/or instructor-provided highlighting (four studies included both type of experiments) (see Table 3). Additionally, 9 studies included the use of highlighting in combination with an additional activity such as training on how to use highlighting, reviewing the highlighted material after some time, or combining highlighting with another learning strategy such as notetaking or graphic organizers (see Table 4); one of these studies was also included in Table 3. Additionally, some treatment groups were analyzed independently since the objective was to observe the effects of irrelevant or inappropriate highlighting on learning; 4 studies were included in this category (see Table 5), and all were parts of studies identified in Table 3.

Data Feature Coding

Each experimental treatment described in the studies was coded using the following descriptor variables: experimental design (i.e., learner-generated highlighting or instructor-provided highlighting), learning outcome measure (i.e., memory or comprehension), and educational level (i.e., college or school).

To compute effect sizes, we used posttest data (sample size, mean, and standard deviation) for the control group and each experimental group (some studies examined more than one type of highlighting). For the highlighting plus another learning activity, we coded the additional activity such as highlighting plus notetaking or training on highlighting plus highlighting.

Effect Size Calculation and Statistical Analyses

Effect sizes were computed using Cohen's *d* (Cohen. 1988) and transformed into Hedge's *g*. We computed the standardized mean difference (Cohen's d) separately for each learning outcome measure and each type of experimental implementation of the highlighting strategy for each study. Thus, more than one effect size can be computed from one publication. When means, standard deviations, and sample sizes were reported for the control and experimental groups, the effect size calculation was as follows—the difference between the means divided by the pooled standard deviations. If only the t value or F value and sample sizes were reported, then we

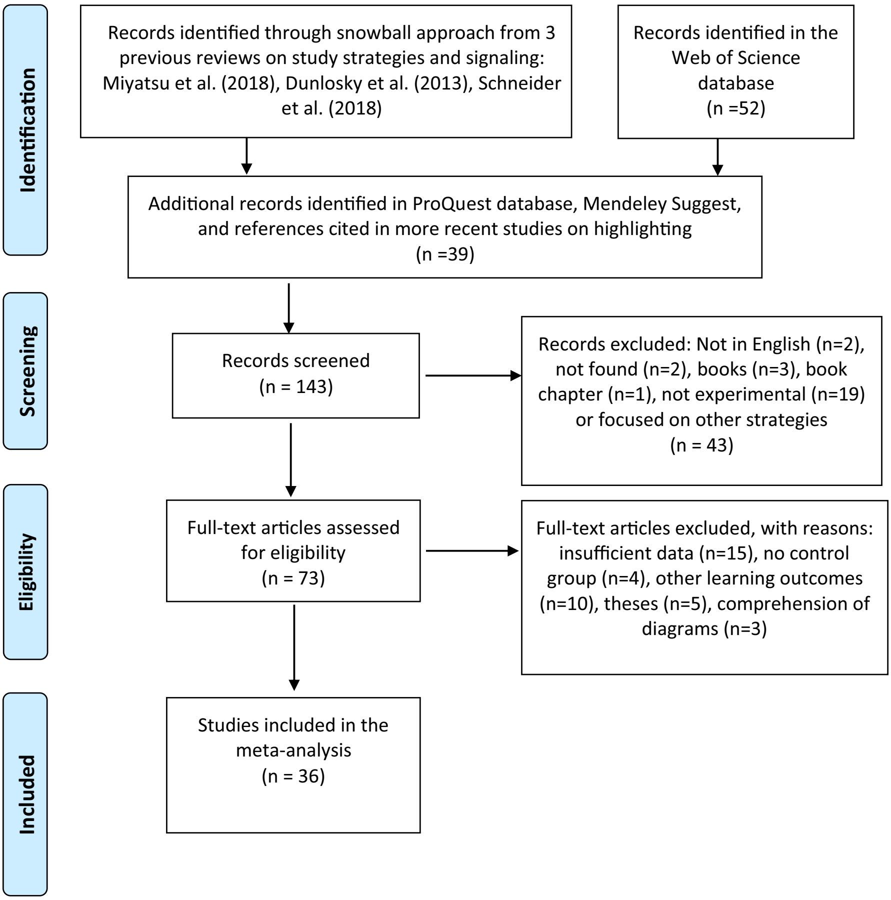

Fig. 3 Phases of Literature Search Strategies

Table 3 Studies Included in the Meta-Analysis on Learner-Generated Highlighting and Instructor-Provided Highlighting

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Mathews (1938)	Learner	School	Underlining	Students underlined important ideas and optionally made marginal notes	Memory	251	245	-0.16	0.09
					Comprehension	251	245	-0.22	0.09
Klare et al. (1955)	Instructor	College	Underlining	Words used in questions asked in the test underlined	Memory	110	110	0.07	0.13
	Instructor	College	Underlining	Important ideas underlined	Memory	110	110	0.21	0.14
Stordahl and Christensen (1956)	Learner	College	Underlining	Subjects underlined main ideas	Comprehension	49	51	-0.04	0.2
Cashen and Leicht (1970)	Instructor	College	Underlining	Selected statements underlined	Memory	20	20	0.64	0.31
Crouse and Idstein (1972)	Instructor	College	Underlining	Subjects were told that underlined passages were the answers to the test	Memory	36	36	0.38	0.24
	Instructor	College	Underlining	Subjects were told to think about questions answered by the underlined passages	Memory	36	36	0.40	0.24
	Instructor	College	Underlining	Subjects read the questions of the test together with the underlined passages	Memory	36	36	0.48	0.24
	Instructor	College	Underlining (Exp. 2)	Subjects were told that the underlined passages contained the answers to the questions in the test	Memory	33	33	3.22	0.38
Leicht and Cashen (1972)	Instructor	College	Underlining	Subjects read a scientific text with scientific principles underlined	Memory	41	41	0.07	0.22
	Instructor	College	Underlining	Same text but with examples underlined	Memory	41	41	0.09	0.22

Table 3 (continued)

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Hoon (1974)	Learner	College	Underlining	Subjects asked to underline important ideas	Comprehension	10	10	0.70	0.44
Schnell and Rocchio (1974)	Instructor	School	Underlining	Main ideas underlined	Memory	26	23	0.91	0.30
	Learner	School	Underlining	Subjects underlined important ideas	Memory	26	23	1.07	0.31
Coles and Foster (1975)	Instructor	College	Underlining (Exp. 1)	Core sentences underlined	Memory	10	10	0.30	0.43
	Instructor	College	Underlining (Exp. 2)	Core sentences underlined + instructions "the important points have been underlined"	Memory	18	20	0.23	0.32
	Instructor	College	Underlining (Exp. 3)	Core sentences underlined + instructions based on SRQ3 method	Memory	20	21	0.53	0.31
Kulhavy et al. (1975)	Learner	School	Underlining	Students asked to underline up to three lines on each page read	Memory	48	48	-0.10	0.20
Rickards and August (1975)	Instructor	College	Underlining	Sentences of high structural importance underlined	Memory	15	15	0.25	0.36
	Learner	College	Underlining	Students underlined sentences of high structural importance	Memory	15	15	0.68	0.37
	Learner	College	Underlining	Students underlined any sentence	Memory	15	15	2.08	0.45
Bialick, Bialick, and Wark (1977)	Instructor	School	Underlining	Main ideas underlined	Memory	24	24	0.90	0.30

Table 3 (continued)

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Fass and Schumacher (1978)	Learner	College	Underlining	Students were asked to underline important ideas. Motivation and text readability controlled	Memory	80	80	0.71	0.23
Rickards and Denner (1979)	Instructor	School	Underlining	Topic sentences underlined	Memory	12	11	-0.49	0.4
	Learner	School	Underlining	Students underlined topic sentences	Memory	12	11	-0.18	0.4
Hartley et al. (1980)	Instructor	School	Underlining	Keywords underlined	Memory	26	29	0.72	0.28
Meyer et al. (1980)	Instructor	School	Underlining	Words that signaled text organizational structure underlined	Memory	64	63	0.36	0.18
Johnson (1988)	Learner	College	Underlining	Students underlined one important sentence per paragraph	Memory	22	24	0.09	0.29
	Learner	College	Underlining (Exp. 2)	Students underlined one important sentence per paragraph and reread the underlined material	Memory	26	23	-0.04	0.29
Golding and Fowler (1992)	Instructor	College	Underlining	Detailed phrases underlined + test expectations controlled	Memory	54	54	0.61	0.28

Table 3 (continued)

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Lorch Jr., Lorch, and Klusewitz (1995)	Instructor	College	Underlining	Target sentences non-underlined, but the rest of the text heavily underlined	Memory	40	42	-0.01	0.22
	Instructor	College	Underlining	Target sentences heavily underlined	Memory	40	42	0.14	0.22
	Instructor	College	Underlining	Target sentences non-underlined, but rest of text lightly underlined	Memory	40	42	0.33	0.22
	Instructor	College	Underlining	Target sentences lightly underlined	Memory	40	42	1.03	0.24
	Instructor	College	Other cues (Exp. 2)	Selected sentences with light capitalization	Memory	38	38	0.48	0.23
Silvers and Kreiner (1997)	Instructor	College	Highlighting	Relevant ideas highlighted	Comprehension	38	38	0.12	0.23
	Instructor	College	Highlighting (Exp. 2)	Relevant ideas highlighted	Comprehension	37	37	0.23	0.23
De Ridder (2000)	Instructor	College	Highlighting	Key vocabulary highlighted	Memory	17	17	0.57	0.34
					Comprehension	17	17	0.46	0.34

Table 3 (continued)

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Simard (2009)	Instructor	School	Underlining	Plural markers underlined	Memory	25	25	0.14	0.28
	Instructor	School	Highlighting	Plural markers highlighted with color only	Memory	25	21	0.45	0.29
	Instructor	School	Other cues	Plural markers enhanced with italic only	Memory	25	29	0.41	0.28
	Instructor	School	Other cues	Plural markers enhanced with bold only	Memory	25	21	0.28	0.29
	Instructor	School	Other cues	Plural markers capitalized only	Memory	25	25	0.68	0.29
	Instructor	School	Other cues	Plural markers enhanced with the use of three typographical cues, that is bold, capital, and underlined	Memory	25	21	0.62	0.29
	Instructor	School	Other cues	Plural markers enhanced with the preceding five typographical cues used at the same time	Memory	25	21	-0.04	0.29
Mahdavi and Azimi (2012)	Learner	School	Underlining	Students underlined important ideas	Comprehension	20	20	0.91	0.32
Ponce and Mayer (2014a)	Learner	College	Highlighting	Students highlighted main ideas	Memory	26	26	0.86	0.29
					Comprehension	26	26	0.36	0.28
	Instructor	College	Highlighting	Main ideas highlighted	Memory	26	26	0.78	0.29
					Comprehension	26	26	0.53	0.28

Table 3 (continued)

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Yue et al. (2014)	Learner	College	Highlighting	Students highlighted relevant ideas and re-studied the highlighted text after 30-min	Memory	46	46	0.19	0.21
	Learner	College	Highlighting	Students highlighted relevant ideas and re-studied the highlighted text immediately	Memory	46	46	0.60	0.21
Ponce et al., (2018a)	Learner	School	Highlighting	Students highlighted words or phrases that signaled text structure	Memory	27	24	0.13	0.28
					Comprehension	27	24	0.15	0.28
Ben-Yehudah and Eshet-Alkalai (2018)	Learner	College	Highlighting	Students highlighted relevant ideas in a digital text	Comprehension	25	25	-0.07	0.28
	Learner	College	Highlighting	Students highlighted relevant ideas in a printed text	Comprehension	26	26	0.48	0.28
Lege (2019)	Instructor	College	Other cues	Text enhanced with various typographical cues and hierarchy	Memory	22	23	0.77	0.3
					Comprehension	22	23	1.05	0.31

Table 4 Studies Included in the Meta-Analysis on Highlighting Combined with other Learning Activities

Authors	Design	Educational level	Strategies	Treatment	Test	Control N	Exp N	*g*	*SE*
Idstein and Jenkins (1972)	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed the underlined material for 4.5 min	Memory	20	20	0.31	0.31
	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed the underlined material for 9 min	Memory	17	19	-0.60	0.33
	Learner	College	Underlining + Reviewing (Exp. 2)	Students underlined text. One week later, they reviewed the underlined material for 15 min	Memory	31	32	0.28	0.25
Schnell and Rocchio (1974)	Learner	School	Training + Underlining	Students received 110 min of training in underlining. Later they underlined text	Memory	26	24	0.86	0.30
Annis and Davis (1978)	Instructor	College	Underlining + Reviewing	Students read underlined text. One week later, they reviewed the underlined material for 10 min	Comprehension	32	32	0.54	0.25
Rickards and Denner (1979)	Instructor	School	Underlining + Post-questions	One sentence was underlined in each paragraph. Each paragraph was followed by content post-questions	Memory	11	12	-0.87	0.42
	Learner	School	Underlining + Post-questions	Students underlined one sentence in each paragraph. Each paragraph was followed by content post-questions	Memory	11	10	0.25	0.42
Peterson (1991)	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed the underlined material for 15 min	Memory	22	22	-0.35	0.29
	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed a non-underlined copy of the material for 15 min	Memory	22	22	-0.24	0.29
	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed the underlined material for 15 min, and a test was applied two months later	Memory	22	22	-0.17	0.29
	Learner	College	Underlining + Reviewing	Students underlined text. One week later, they reviewed a non-underlined copy of the material for 15 min, and a test was applied two months later	Memory	22	22	-0.02	0.29

Table 4 (continued)

Authors	Design	Educational level	Strategies	Treatment	Test	Control N	Exp N	g	SE
Amer (1994)	Learner	College	Training + Underlining	Students received five 90-min sessions, one session per week, of training in underlining. During the experiment, students underlined the text	Comprehension	31	33	0.66	0.26
Leutner, Leopold, and Den Elzen-Rump (2007)	Learner	College	Training + Highlighting	Students received three sessions for 93 min of training on highlighting. During the experiment, students applied this knowledge	Comprehension	15	15	0.57	0.36
	Learner	College	Training + Highlighting + Self-regulation	Students received three sessions that lasted 93 min of training on highlighting and self-regulation. During the experiment, students applied this knowledge	Comprehension	15	15	1.13	0.38
Hayati and Shariatifar (2009)	Learner	College	Training + Underlining	Students received 60-min training. During the experiment, students underlined important words, ideas, and supporting details in each paragraph	Comprehension	20	20	2.12	0.39
Ponce et al., (2018a)	Learner	School	Highlighting + Notetaking	Students highlighted text and took written notes	Memory	27	24	0.82	0.29
					Comprehension	27	24	0.74	0.29
	Learner	School	Highlighting + Graphic Organizer	Students highlighted text and filled in a graphic organizer	Memory	27	24	0.92	0.30
					Comprehension	27	24	1.33	0.31

used these data to compute effect sizes. If the F value was reported for ANOVA with three or more groups, and the means and sample sizes for each group were reported, we used the procedure described by Lipsey and Wilson (2001) to compute the effect size for each possible comparison between control and experimental groups. To compute Hedge's g and its standard error (SE), we used the following formulas (Borenstein et al. 2009): $g = J \text{ x } d$, $V_g = J^2 \text{ x } V_d$, and $SE_g = \text{sqrt}(V_g)$, where $J = (1\text{-}(3/(4(n_1 + n_2 - 2) - 1)))$, and n_1 and n_2 correspond to the sample sizes of the control and experimental groups.

Several studies reported data from two or more independent treatment groups. For example, Simard (2009) experimented with different types of typographical cues for highlighting which included one control group and seven treatment subgroups. We followed Borenstein et al.'s (2009) recommendations and computed a composite effect size for each study by performing a fixed-effect meta-analysis on the subgroups, where the mean effect size is the sum of the products of each effect size multiplied by its weight (i.e., the inverse of its variance) divided by the sum of the weights. The variance was computed as the reciprocal of the sum of the weights and the standard error as the square root of the variance. The following studies were treated in this way (see Table 1): Klare et al. (1955), Crouse and Idstein (1972) (Exp. 1 only), Leicht and Cashen (1972), Rickards and August (1975), Lorch Jr. et al. 1995) (Exp. 1 only), and Simard (2009). Additionally, to compute average effect sizes for educational level (i.e., college and K-12), we computed a combined effect size for each study or experiment within the study that reported memory and comprehension test scores for the same participants. The following procedure was used (Borenstein et al. 2009): First, the average between effect sizes for memory and comprehension was computed and second, the new variance was computed as the sum of both variances plus twice the correlation between memory and compression scores by the square roots of each variance; and the result was divided by four. We assumed a 0.5 correlation between memory and comprehension scores for all studies in this condition (De Ridder. 2000; Lege. 2019; Mathews. 1938; Ponce & Mayer. 2014a; Ponce et al. 2018a).

A random-effects analysis method with a restricted maximum likelihood estimator (REML) was used to carry out the meta-analyses (Borenstein et al. 2009). The quantity I^2 or level of heterogeneity was used to assess the consistency of the meta-analyses' results. Deeks et al. 2020) offer the following rough guides for the interpretation of the I^2 statistic: (1) 0% to 40% might not be significant; (2) 30% to 60% may represent moderate heterogeneity; (3) 50% to 90% may represent substantial heterogeneity; and (4) 75% to 100% considerable heterogeneity. The authors argue that the interpretation of this statistic depends on several factors such as magnitude and direction of effects, number of studies included in the meta-analysis, and strength of evidence for heterogeneity.

Prediction intervals were used to estimate the dispersion of effect sizes. Borenstein (2019) indicates that confidence interval and prediction interval are different types of statistical indices. The confidence interval shows the precision of the estimated mean effect, whereas the prediction interval shows the dispersion of effects or how broadly the effect size differs across populations. A

Table 5 Inappropriate Text Highlighting

Authors	Design	Educational Level	Typographical cue	Treatment	Test	Control N	Exp N	*g*	*SE*
Leicht and Cashen (1972)	Instructor	College	Underlining	Trivial statements underlined	Memory	41	41	0.22	0.22
Rickards and August (1975)	Learner	College	Underlining	Students underlined statements of low structural importance	Memory	15	15	-0.56	0.36
	Instructor	College	Underlining	Statements of low structural importance underlined	Memory	15	15	0.20	0.36
Bialick, Bialick, and Wark (1977)	Instructor	School	Underlining	Inappropriate underlining	Memory	24	24	-0.93	0.30
Silvers and Kreiner (1997)	Instructor	College	Highlighting	Inappropriate highlighting	Comprehension	38	38	-2.05	0.28
	Instructor	College	Highlighting (Exp. 2)	Students warned of inappropriate highlighting	Comprehension	37	37	-1.07	0.25

95% prediction interval will tell us that the true effect size will be as low as the lower point of the interval for some populations and as high as the higher point on other populations. Borenstein (2019) provided the formula and accompanied software application to compute prediction intervals.

Outliers or influential cases were analyzed to observe their impacts not only on the average effect sizes computed but also on their effects on the heterogeneity of the results. As indicated by Deeks et al. (2020), one or two studies that conflict with the rest of the studies can generate heterogeneity. Effect sizes were transformed to Z-scores; outliers with a Z value lower than -3.3 or larger than 3.3 were windsorized by adjusting their values to the next-lowest or next-largest effect size in the distribution, as used in previous meta-analyses (Castro-Alonso et al. 2019).

We used the JASP software (JASP-TEAM. 2020) to carry out each of the meta-analyses included in this study. Moderator analyses were conducted for learning outcome and educational level measures using subgroup analyses with a mixed-effects model and a pool within-group estimates of tau-squared (Borenstein. 2019). The subgroup analyses and publication bias analysis were performed with the Meta-Essentials tool for meta-analysis (Suurmond et al. 2017).

Results

A total of 85 effect sizes were generated from the data recollected from the 36 articles, as shown in Tables 3, 4, and 5. Sixty effect sizes were associated with learner-generated highlighting and instructor-provided highlighting (see Table 3), which were reduced to 46 effect sizes after applying the procedure indicated in the methods section to the six studies that included more than one treatment conditions. Additionally, nineteen effect sizes were related to experimental treatments that combined highlighting with another learning activity (see Table 4), and six effect sizes were computed from experimental treatments that examined the effects of inappropriate text highlighting (see Table 5).

To test our hypotheses, we conducted independent meta-analyses with these effect sizes separated by learner-generated highlighting or instructor-provided highlighting as presented in Tables 6, 7, and 8. The decision to conduct independent meta-analyses was based on the hypotheses stated and the objective of this study—to identify under what conditions highlighting support learning.

The reporting of the results of the meta-analyses was based on Borenstein's (2019) recommendations. We will use Hattie's (2009) meta-analysis findings for interpreting effect sizes in education, indicating that an effect size Cohen's $d = 0.40$ is the "hinge point" at which an educational intervention shows a greater than average impact on student achievement.

Tables 6, 7, and 8 present the results of the meta-analyses. Each table includes the total number of participants (*N*) in each category, the number of effect sizes computed (*k*), the weighted effect size ($g+$), its standard error (*SE*), and its p value (*p*), the 95% confidence interval (CI), the results of the test of heterogeneity (*Q*), its p value (p), and the percentage of heterogeneity (I^2), the between-study variance (T^2), and the 95% prediction interval (PI).

Learner-generated highlighting

Table 6 presents the results of the meta-analysis for learner-generated highlighting. One outlier was identified for the learner-generated highlighting corresponding to the second experiment in the study of Crouse and Idstein (1972), with $g = 3.22$ (Z value > 3.3). We windsorized this outlier to the next-largest value ($g = 1.03$), considering learning outcome and educational level. Table 7 presents the results for learner-generated highlighting with added features such as using highlighting in combination with another strategy.

Does learner-generated highlighting improve learning?

Hypothesis 1 predicts that when students use highlighting as learning strategy, they will show improvements in tests of memory but not in tests of comprehension. The second row of Table 6 presents the standardized mean difference for tests of memory, with $g+ = 0.36$ and confidence interval between 0.08 and 0.64. The Z value for testing the null hypothesis was 2.489 with $p = 0.013$. Consequently, on average, learners who used highlighting as learning strategy (experimental groups) improved memory (e.g., recall of main ideas) by 0.36 standard deviations compared to learners who only read or studied the text (control groups). This mean effect size is within Hattie's "hinge point" range, indicating that, on average, highlighting as a learning strategy has a significant and relevant impact on supporting students to remember important ideas in the learning material.

Effect size variation across studies, $Q(1,11) = 54.3$, $p < 0.001$, indicates that the actual effect size is not identical in all the studies. The I^2 statistic equal to 79% reflects a substantial level of heterogeneity among the studies. The T^2 statistics was 0.178, and the corresponding standard deviation T was 0.421. The 95% prediction interval was -0.63 to 1.35, which tells us that in 95% of all populations comparable to the analysis, the actual effect will fall in this range. It reflects the uncertainty in the overall effect size if a new study is included in the meta-analysis. For example, some studies included in this meta-analysis found that highlighting as learning strategy is not effective (reflected in negative effect sizes) (Mathews. 1938; Rickards & Denner. 1979) but other studies found that highlighting is a very effective learning strategy (reflected in positive and large effect sizes) (Fass & Schumacher. 1978; Rickards & August. 1975), so it is expected that new studies will fall within the prediction interval.

Hypothesis 1 also predicts that when students use highlighting as learning strategy, they will not show improvements in comprehension tests. The third row of Table 6 presents the standardized mean difference in tests of comprehension, with $g+ = 0.20$, confidence interval between -0.08 and 0.48, and Z value equal to 1.421 ($p = 0.155$). This implies that, on average, learners who used highlighting as a learning strategy did not show improvements in comprehension compared to learners who

Table 6 Weighted Mean Effect Size for Learner-Generated Highlighting

Category	N	k	Effect size			95% CI		Test of heterogeneity			T^2	95% PI	
			$g+$	SE	p	LCI	UCI	Q	p	I^2 (%)		LPI	UPI
Learning outcome													
Memory	1251	12	0.36	0.14	.013	0.08	0.64	54.3	< .001	79	0.178	-0.63	1.35
Comprehension	861	8	0.20	0.14	.155	-0.08	0.48	21.5	.003	65	0.094	-0.62	1.03
Educational level													
College	758	11	0.39	0.12	.001	0.15	0.63	24.6	.006	60	0.097	-0.37	1.15
School	755	6	0.24	0.22	.278	-0.20	0.68	25.9	< .001	84	0.232	-1.24	1.72

Table 7 Weighted Mean Effect Size for Learner-Generated Highlighting with Added Features

Category	N	k	Effect size			95% CI		Test of heterogeneity			T^2	95% PI	
			$g+$	SE	p	LCI	UCI	Q	p	I^2 (%)		LPI	UPI
Added Features													
Training	184	4	1.02	0.34	.002	0.36	1.69	11.4	.010	77	0.349	-1.91	3.95
Notetaking, graphic organizer, or post-questions	123	3	0.82	0.21	< .001	0.40	1.24	3.3	.188	35	0.049	-3.09	4.73
Review	271	5	-0.06	0.16	.692	-0.38	0.25	6.8	.146	40	0.051	-0.94	0.82

Table 8 Weighted Mean Effect Size for Instructor-Provided Highlighting

Category	N	k	Effect size			95% CI		Test of heterogeneity			T^2	95% PI	
			$g+$	SE	p	LCI	UCI	Q	p	I^2 (%)		LPI	UPI
Learning outcome													
Memory	1845	21	0.44	0.07	< .001	0.32	0.57	35.2	.019	45	0.033	0.04	0.84
Comprehension	281	5	0.44	0.16	.006	0.13	0.75	6.7	.154	40	0.049	-0.43	1.30
Educational level													
College	1505	17	0.41	0.07	< .001	0.27	0.54	26.0	.055	43	0.030	0.01	0.81
School	490	6	0.48	0.15	.002	0.18	0.78	11.9	.036	61	0.077	-0.40	1.36

only read or studied the text. The I^2 statistic equal to 65% reflects a substantial level of heterogeneity among the studies, and the prediction interval between -0.62 and 1.03.

Additionally, we computed the overall effect size for highlighting as learning strategy (memory and comprehension were treated separately), resulting in $g+=0.30$, confidence interval between 0.10 and 0.50 (Z value = 2.944, $p=0.003$). We also examined learning outcome measure as a moderator. The between-levels difference was not statistically significant, $Q_B(1)=0.3$, $p=0.561$, signaling that highlighting was beneficial for learning independently of the test used to measure learning outcomes.

Does learner-generated highlighting improve learning for college and K-12 students?

Hypothesis 2 posits that college students who use highlighting as a learning strategy will show improvements in learning outcomes. The sixth row of Table 6 presents the standardized mean difference for college students, with $g+=0.39$ and confidence interval between 0.15 and 0.63 (Z = 3.176, $p=0.001$). This effect size confirms that, on average, college students who used highlighting as a learning strategy improved learning significantly more than college students who only read or studied the text. A significant effect size variation across studies was found, $Q(1,10)=24.6$, $p=0.006$, with $I^2=60\%$ and a prediction interval between -0.37 and 1.15.

Hypothesis 2 also posits that K-12 students do not benefit from using highlighting as a learning strategy. The seventh row of Table 6 presents the standardized mean difference $g+=0.24$, with a confidence interval between -0.20 and 0.68 (Z = 1.085, $p=0.278$). This effect size confirms that school students, on average, did not improve learning when using highlighting as a learning strategy. The results also reveal a substantial level of heterogeneity, $I^2=84\%$, and a prediction interval between -1.24 and 1.72.

We examined educational level as moderator over the combined effect size for highlighting as learning strategy (memory and comprehension were combined in one learning measure, as indicated in the method section). The overall effect size was $g+=0.33$ and confidence interval between 0.12 and 0.55 (Z value = 3.005, $p=0.003$), and the between-levels difference was not statistically significant, $Q_B(1)=0.4$, $p=0.517$, signaling that highlighting was beneficial for learning independent of the educational level of the students.

Does learner-generated highlighting improve learning after receiving training?

Hypothesis 3 proposes that students who receive training on how to highlight show improvements in learning outcomes. The first row of Table 6 shows that the standardized mean difference is $g+=1.02$, with a confidence interval between 0.36 and 1.69 (Z = 3.022, $p=0.002$). This effect size confirms that, on average, students who received training on highlighting before using highlighting as learning strategy improved learning significantly more than college students who only read or studied the text. A significant effect size variation across studies was found, $Q(1,3)=11.4$, $p=0.010$, with $I^2=77\%$ and a prediction interval between -1.91 and 3.95.

Does learner-generated highlighting improve learning when used with a complementary learning strategy?

Hypothesis 4 holds that learner-generated highlighting yields improvements in learning outcomes when a complementary learning strategy is added (such as notetaking or constructing a graphic organizer). The second row of Table 7 shows that the standardized mean difference is $g+=0.82$, with a confidence interval between 0.40 and 1.24 ($Z=3.841$, $p<0.001$). This mean effect size confirms that, on average, students who used highlighting with a complementary learning strategy improved learning significantly more than students who only read or studied the text. A nonsignificant effect size variation across studies was found, $Q(1, 2)=3.4$, $p=0.186$, with $I^2=36\%$ and a prediction interval between -3.09 and 4.73. These results show the high consistency across the studies, and the effectiveness of combining highlighting with notetaking or constructing a graphic organizer, although the number of studies is still low.

Does learner-generated highlighting improve learning when asked to review what they had highlighted?

Hypothesis 5 posits that when students are asked to review material they have highlighted previously (e.g., one week), learning does not improve. The final row of Table 7 shows that the standardized mean difference is $g+=-0.06$, with a confidence interval between -0.38 and 0.25 ($Z=-0.396$, $p=0.692$). This effect size confirms that, on average, reviewing highlighted text previously highlighted by learners did not improve learning significantly more than students who only read or studied the text. A nonsignificant effect size variation across studies was found, $Q(1,4)=6.8$, $p=0.146$, with $I^2=40\%$ and a prediction interval between -0.94 and 0.82, which show high consistency across the studies, although with a dispersion of the effect sizes signaled by the prediction interval.

Instructor-provided highlighting

Table 8 presents the results of the meta-analysis for instructor-provided highlighting derived from effect sizes included in Tables 3 and 4 for this category. No outliers were identified for instructor-provided highlighting (all effect sizes transformed to Z-scores in this category were not greater than 3.3 or less than -3.3).

Does instructor-provided highlighting improve learning?

Hypothesis 6 is that when students study highlighted text, they will show improvements in tests of memory and comprehension. The second row of Table 8 shows that the standardized mean difference for tests of memory is $g+=0.44$, with a confidence interval between 0.32 and 0.57. The Z value for testing the null hypothesis was 6.843 with $p < 0.001$. Regarding comprehension tests, the third row of Table 8 shows that the standardized mean difference is $g+ = 0.44$, confidence interval between 0.13 and 0.75, and Z value equal to 2.768, with $p = 0.006$. Consequently, on average, learners who received high-quality highlighting (experimental groups) improved learning by 0.44 standard deviations on memory and comprehension tests, respectively, compared to learners who only read or studied the text (control groups). These mean effect sizes are within Hattie's "hinge point" range. The levels of heterogeneity and prediction intervals for memory and comprehension support the consistency of these findings.

We also examined learning outcome measure as a moderator over the overall effect size for instructor-provided highlighting (memory and comprehension were treated separately). The overall effect for instructor-provided highlighting was $g+ = 0.44$, confidence interval between 0.32 and 0.55 (Z value = 7.511, $p = < 0.001$). The between-levels difference was not statistically significant, $Q_B(1) = 0.0$, $p = 0.934$, showing that highlighting was beneficial for learning independent of the test used to measure learning outcomes.

Does instructor-provided highlighting improve learning for college and K-12 students?

Hypothesis 7 is that college and K-12 students who receive instructor-provided highlighting improve learning. The fifth row of Table 8 shows that the standardized mean difference for college students is $g+ = 0.41$, a confidence interval between 0.27 and 0.54 (Z = 5.820, $p < 0.001$). Regarding K-12 students, the standardized mean difference is $g+ = 0.48$, with a confidence interval between 0.18 and 0.78 (Z = 3.158, $p = 0.002$). These effects sizes confirm that, on average, college and school students who are provided with high-quality highlighting improved learning more than students who only read or studied the text. The levels of heterogeneity and prediction intervals support the consistency of these findings.

We additionally examined educational level as moderator over the overall effect size for highlighting as learning strategy (memory and comprehension were combined in one learning measure, as indicated in the method section). The overall effect for instructor-provided highlighting was $g+ = 0.42$, confidence interval between 0.30 and 0.54 (Z value = 7.007, $p = < 0.001$). The between-levels difference was not statistically significant, $Q_B(1) = 0.2$, $p = 0.640$, signaling that highlighting was beneficial for learning independent of the educational level of the students.

Does inappropriate highlighting hurt learning?

We identified a few experimental treatments in which the objective was to examine the effects of inappropriate text highlighting in tests of memory or comprehension. These studies are presented in Table 5. The overall average effect size computed was $g+ = -0.70$ ($NE = 6$, Z = -1.987, $p = 0.047$, 95% CI [-1.40, -0.01]) and considerable level of heterogeneity observed among the studies, $Q(1, 5) = 49.9$, $p < 0.001$, $I^2 = 89\%$, and a 95% prediction interval between -3.21 and 1.78. On average, as expected, inappropriate highlighting is detrimental to learning.

Examining publication bias

Additionally, we investigated the possibility of publication bias because research with significant findings tend to be published more frequently than research with nonsignificant findings. In this examination, studies included in the learner-generated highlighting section and instructor-provided highlighting section in which memory and comprehension were combined in one learning measure were considered (i.e., college and K-12 analyses). The Trim and Fill method tells us that the plot is symmetrical as shown in Fig. 4, which indicates that publication bias is not likely to be present in this set of studies.

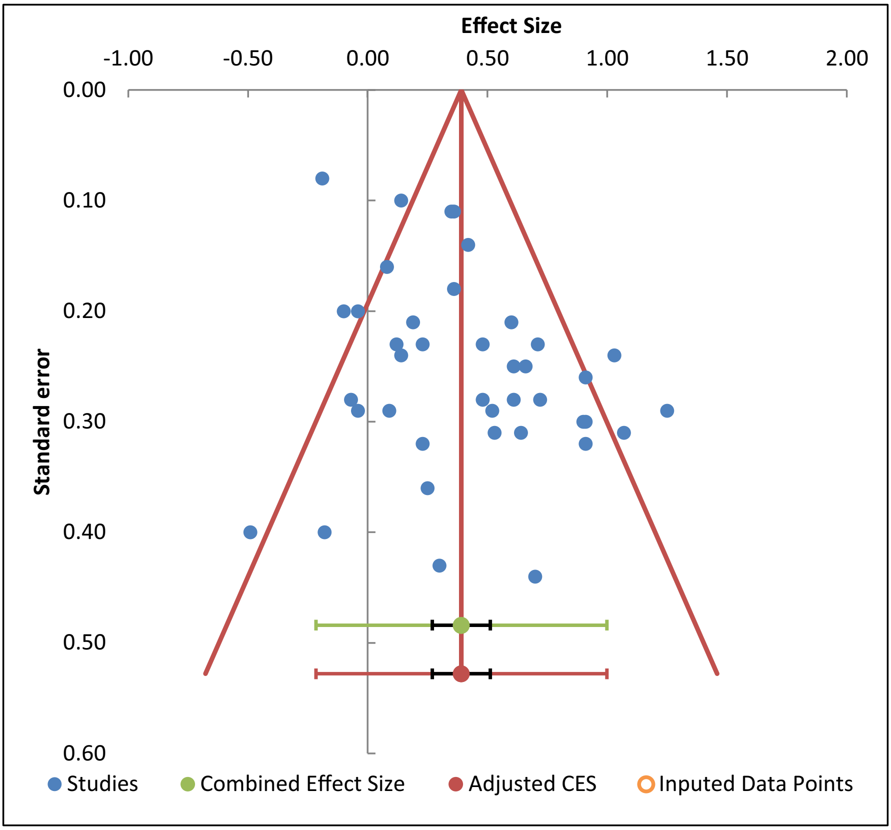

Fig. 4 Funnel plot

Discussion

Empirical Contribution

In this study, we examined learner-generated highlighting as a learning strategy and instructor-provided highlighting as an instructional design approach. The results from the meta-analyses show that, on average, when learners generated their own highlighting, they improved on tests of memory but not on tests of comprehension, with average effect sizes of 0.36 and 0.20, respectively (Hypothesis 1). When learners received high-quality highlighting, they improved on tests of memory and comprehension, both with an average effect size of 0.44 (Hypothesis 6). Based on Hattie's (2009) interpretation of effect sizes in education, these are significant findings; it can be concluded that, on average, highlighting shows a greater than average impact on student learning, particularly when it is implemented as an instructional design feature.

The results from the meta-analyses also demonstrated that when college students used highlighting as a learning strategy, they benefited significantly but K-12 students did not, with effect sizes of 0.39 and 0.24, respectively (Hypothesis 2). These findings demonstrate that less experienced learners may not know what to highlight, so training will be more necessary for younger students. Rickards and Denner (1979) argue that children compared with more mature students are still developing meta-comprehension skills so such deficiency may lead them to select details rather than main ideas to highlight. In a more recent review, Miyatsu et al. (2018) also have noted that students who are younger than college age generally do not benefit from learner-generated highlighting. Studies with K-12 students on highlighting have employed expository texts; however, children are more familiar with narrative texts, which present important differences with expository texts not only in terms of content but also in their organizational structure. Such unfamiliarity with text structure is an important limitation for the appropriate use of highlighting as learning strategy (Meyer & Poon. 2001; Ponce et al. 2018a).

Regarding instructor-provided highlighting, this meta-analysis reveals that when high-quality highlighting is provided, on average, both college and K-12 students benefit greatly, with effect sizes of 0.41 and 0.48, respectively (Hypothesis 7). Training on how to use highlighting as a learning strategy showed encouraging results. Despite the small number of studies, it shows that training has a significant and higher than average effect size on learning from text, with an average effect size of 1.02 (Hypothesis 3). As one of the studies demonstrated (Schnell & Rocchio. 1974), even less than 2 h of training can have a considerable effect on learning from texts ($g=0.86$).

The results from combining highlighting with another learning strategy such as notetaking or graphic organizers were demonstrated to be very effective, with an average effect size of 0.82 (Hypothesis 4). This average effect size represents more than twice the largest average effect sizes observed for highlighting as single learning strategy (i.e., college students, $g+=0.39$). When highlighting is used in conjunction with another learning strategy such as notetaking, graphic organizer, or post-question, the effectiveness of such combinations is particularly high, as demonstrated by the few studies in this area.

For example, Ponce et al.'s (2018a) study showed that asking students, first, to highlight text, and second, to take notes or fill-in a graphic organizer is much more effective than asking students to either read the text or highlight relevant ideas in the text.

However, asking students to review their highlighting after some time was not more effective than reading the learning material, with an average effect size of -0.06 (Hypothesis 5). The experimental studies examined in this meta-analysis asked students to review the highlighted material one week after the students highlighted such material. The main variations were in the time spent reviewing or studying the highlighting texts, ranging from 4.5 min to 15 min (see Table 4 for more details). Failure to highlight relevant information (e.g., main ideas) from text can lead students to review their own inappropriate or irrelevant study material (Peterson. 1991).

Finally, experimental treatments designed to examine the effects of inappropriate highlighting demonstrated the detrimental consequences of this type of highlighting on learning, with an average effect size of -0.70. This finding informs instructional designers about the appropriate use of the highlighting technique when developing instructional material to support learning, such as choosing to highlight main ideas and/or organizational elements in texts.

Theoretical Contributions

This study was grounded in generative learning theory (Fiorella & Mayer. 2015; Wittrock. 1989) and the cognitive theory of multimedia learning (Mayer. 2021) which postulate that active learning occurs when learners engage in appropriate cognitive processing during learning. As predicted by these theories, without training on what to highlight, learner-generated highlighting activates mainly the cognitive process of selecting. This was confirmed by this meta-analysis; on average, learner-generated highlighting showed improvements in memory tests but not in comprehension tests. It demonstrates that the type of learning that emerges is mainly confined to superficial learning in which learners tend to memorize a list of facts rather than a deeper elaboration of the learning material. Similarly, this meta-analysis supports the prediction that learner-generated highlighting works mainly for college students. K-12 students may not know what to highlight or have less experience with this technique, such as not knowing how to distinguish main ideas from secondary ideas or how to identify the organizational structure of text (e.g., cause-and-effect). On the hand, when less experienced students receive training on what to highlight, it leads to the activation of selecting, organizing, integrating cognitive processes. Learners learn what to search in a text that needs to be highlighted, activating thus the selecting cognitive process and leaving cognitive capacity available for the higher-level processes of organizing and integrating.

Asking students to highlight text in conjunction with a second strategy, such as constructing a graphic organizer or notetaking, promotes deeper processing of the text that fosters deeper comprehension of the text, as demonstrated by better scores on memory and comprehension. As predicted, the highlighted information is selected and stored in working memory for further processing; these strategies provide structures to organize the selected ideas, allowing the construction of a mental model of the text and integration with previous knowledge.

This meta-analysis also confirms that when learners receive high-quality highlighting, learner's selecting processes are conducted with high efficiency, and therefore, cognitive capacity is available for the higher-level processes of organizing and integrating, as shown by improvements in learning by college and K-12 students in both memory and comprehension measures.

Practical contributions

Instruction on how to use the highlighting strategy can be crucial to improve its impact on learning for less experienced learners. Some recommendations from the reviewed articles in this study are the following. During instruction, instructors should provide demonstration and practice using instructor-based models of effective and non-effective highlighting techniques. More structured and instructor-based methods of highlighting are more beneficial than non-structured or non-instructed approaches. A structured-instruction approach includes discrimination practice on how to differentiate main ideas from secondary or supporting ideas, or how to identify cues in the text that show its organizational structure (e.g., sequence, cause-and-effect, or comparison types of texts), and shows students examples of appropriate and inappropriate highlighting, and offers relevant feedback. One or two hours of instructions have demonstrated sufficient to observe improvement in memory measures (Hayati & Shariatifar. 2009; Leutner et al. 2007; Schnell & Rocchio. 1974). Learners and instructional designers should highlight content relatively sparingly to observe improvements in learning from text (Lorch et al. Jr. 1995). If too much text has been highlighted, learners become desensitized to highlighting, which can harm learners' ability to remember relevant ideas from text (Lege. 2019).

The combined use of highlighting with notetaking, graphic organizers, or self-regulatory cognitive processes has also proved effective in improving student learning (Leutner et al. 2007; Ponce et al. 2018a). Therefore, a practical recommendation could be to teach students how to highlight by identifying relevant information in combination with the use of graphic organizers. In this case, the structure of the graphic organizer (e.g., compare-and-contrast organizer) can be used to identify key information in the text.

Future Research and Limitations

Given that learner-generated highlighting was not very helpful for K-12 students, there is a need to understand other factors that can be moderating the appropriate use of highlighting, such as learner's reading comprehension level, vocabulary knowledge, lack of training on highlighting, and previous training on other reading strategies (e.g., notetaking and questioning). Will continued exposure to high-quality highlighted text improve learner-generated highlighting?

Primary school students involved in the experiments were fifth and sixth graders only. Will younger students benefit from highlighting text or studying highlighted text? Rickards and Denner's (1979) study found detrimental effects with fifth graders; however, the studies by Hartley et al. (1980), Simard (2009), and Ponce et al. (2018a) found positive effects with sixth graders. Are these effects replicable with learners from different grades and school contexts?

Most texts used in the experiments with highlighting were expository texts, focusing on the content of the text (e.g., highlighting main ideas, principles, or details). Previous research indicated that a critical element in helping learners to comprehend expository texts is to work not only on its content but also on its internal organization or structure (e.g., cause-and-effect or compare-and-contrast text types) (Cook & Mayer. 1988; Fiorella & Mayer. 2016; Meyer & Poon. 2001; Ponce & Mayer. 2014b). Consequently, a possible research avenue is to examine whether highlighting helps learners to recognize text structures and at the same time support the comprehension of its content (e.g., highlighting keywords that characterize text structures).

Additionally, narrative texts (e.g., stories and tales) are extensively used in elementary school, and instruction of its content and internal structures (i.e., story grammar) have been shown to improve memory and comprehension of this type of text (National Institute of Child Health & Human Development. 2000). Should instruction on highlighting start with narrative texts before teaching how to highlight expository texts? Are the skills developed to identify story structures transferable to expository text?

More research is also needed to explore the effects of using highlighting in combination with other strategies before, during, and after reading. Some combinations have shown to be effective such as highlighting plus notetaking or highlighting plus graphic organizer (Ponce et al. 2018a), but others have proven less effective, such as instructor-provided highlighting plus content questions (Rickards & Denner. 1979).

Finally, a potential limitation of this study is that we focused on peer-reviewed studies; postgraduate theses and unpublished research studies were not included in this meta-analysis.

Funding This work was supported by the Agencia Nacional de Investigación y Desarrollo de Chile under Grant FONDECYT #1200803.

References

Amer, A. A. (1994). The effect of knowledge-map and underlining training on the reading comprehension of scientific texts. *English for Specific Purposes, 13*(1), 35–45. https://doi.org/10.1016/0889-4906(94)90023-X

Annis, L., & Davis, J. K. (1978). Study techniques and cognitive style: Their effect on recall and recognition. *The Journal of Educational Research, 71*(3), 175–178. https://doi.org/10.1080/00220671.1978.10885064

The contribution of text-highlighting to comprehension: A comparison of print and digital reading. Journal of Educational Multimedia and Hypermedia, 27(2), 153–178.Retrieved from https://www.learntechlib.org/primary/p/174353/

Bialick, S. M., Bialick, D. H., & Wark, D. M. (1977). The effects of experimenter-underlining and expertise on the retention of textual material. In P. D. Pearson & J. Hansen (Eds.), Reading: Theory, Research, and Practice. Twenty-Sixth Yearbook of the National Reading Conference (pp. 26–31). National Reading Conference.Retrieved from https://files.eric.ed.gov/fulltext/ED227440.pdf#page=37

Borenstein, M. (2019). *Common mistakes in meta-analysis and how to avoid them*. Biostat Inc.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. John Wiley & Sons.

Cashen, V. M., & Leicht, K. L. (1970). Role of the isolation effect in a formal educational setting. *Journal of Educational Psychology, 61*(6), 484–486. https://doi.org/10.1037/h0030286

Castro-Alonso, J. C., Wong, M., Adesope, O. O., Ayres, P., & Paas, F. (2019). Gender Imbalance in Instructional Dynamic Versus Static Visualizations: A Meta-analysis. *Educational Psychology Review, 31*, 361–387. https://doi.org/10.1007/s10648-019-09469-1

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). L. Erlbaum Associates.

Coles, P., & Foster, J. (1975). Typographic cueing as an aid to learning from typewritten text. *Innovations in Education & Training International, 12*(2), 102–108. https://doi.org/10.1080/1355800750120206

Cook, L. K., & Mayer, R. E. (1988). Teaching readers about the structure of scientific text. *Journal of Educational Psychology, 80*, 448–456. https://doi.org/10.1037/0022-0663.80.4.448

Crouse, J. H., & Idstein, P. (1972). Effects of encoding cues on prose learning. *Journal of Educational Psychology*, *63*(4), 309–313. Retrieved from https://www.jstor.org/stable/20151221

De Ridder, I. (2000). Are we conditioned to follow links? Highlights in CALL materials and their impact on the reading process. *Computer Assisted Language Learning, 13*(2), 183–195. https://doi.org/10.1076/0958-8221(200004)13:2;1-d;ft183

Deeks, J., Higgins, J., & Altman, D. (2020). Chapter 10: Analysing data and undertaking meta-analyses. In J. Higgins, J. Thomas, J. Chandler, M. Cumpston, T. Li, M. Page, & V. Welch (Eds.), *Cochrane Handbook for Systematic Reviews of Interventions version 6.1 (updated September 2020)*. Cochrane. Retrieved from www.training.cochrane.org/handbook

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest, 14*(1), 4–58. https://doi.org/10.1177/1529100612453266

Fass, W., & Schumacher, G. M. (1978). Effects of motivation, subject activity, and readability on the retention of prose materials. *Journal of Educational Psychology, 70*(5), 803–807. https://doi.org/10.1037/0022-0663.70.5.803

Fiorella, L., & Mayer, R. E. (2015). *Learning as a generative activity*. Cambridge University Press.

Fiorella, L., & Mayer, R. E. (2016). Eight ways to promote generative learning. *Educational Psychology Review, 28*(4), 717–741. https://doi.org/10.1007/s10648-015-9348-9

Goldingand, J. M., & Fowler, S. B. (1992). The limited facilitative effect of typographical signals. *Contemporary Educational Psychology, 17*(2), 13. https://doi.org/10.1016/0361-476X(92)90052-Z

Hartley, J., Bartlett, S., & Branthwaite, A. (1980). Underlining can make a difference sometimes. *The Journal of Educational Research, 73*(4), 218–224. https://doi.org/10.1080/00220671.1980.10885239

Hattie, J. (2009). *Visible learning : A synthesis of over 800 meta analyses relating to achievement*. Routledge.

Hayati, A. M., & Shariatifar, S. (2009). Mapping strategies. *Journal of College Reading and Learning, 39*(2), 53–67. https://doi.org/10.1080/10790195.2009.10850318

Hoon, P. W. (1974). Efficacy of three common study methods. *Psychological Reports, 35*(3), 1057–1058. https://doi.org/10.2466/pr0.1974.35.3.1057

Idstein, P., & Jenkins, J. R. (1972). Underlining versus repetitive reading. *The Journal of Educational Research, 65*(7), 321–323. https://doi.org/10.1080/00220671.1972.10884331

JASP-TEAM. (2020). JASP (Version 0.14.1) Computer Program. Retrieved from https://jasp-stats.org

Johnson, L. L. (1988). Effects of underlining textbook sentences on passage and sentence retention. *Reading Research and Instruction, 28*(1), 18–32. https://doi.org/10.1080/19388078809557955

Klare, G. R., Mabry, J. E., & Gustafson, L. M. (1955). The relationship of style difficulty to immediate retention and to acceptability of technical material. *Journal of Educational Psychology, 46*(5), 287–295. https://doi.org/10.1037/h0044458

Kulhavy, R. W., Dyer, J. W., & Silver, L. (1975). The effects of notetaking and test expectancy on the learning of text material. *Journal of Educational Research, 68*(10), 363–365. https://doi.org/10.1080/00220671.1975.10884802

Lege, R. (2019). Typography's effect on language learners' reading processes: An eye tracking study. *The Asian EFL Journal*, *23*(3.1), 134–153. Retrieved from https://www.researchgate.net/publication/338763475
Leicht, K. L., & Cashen, V. M. (1972). Type of highlighted material and examination performance. *Journal of Educational Research, 65*(7), 315–316. https://doi.org/10.1080/00220671.1972.10884329
Leutner, D., Leopold, C., & Den Elzen-Rump, V. (2007). Self-regulated learning with a text-highlighting strategy a training experiment. *Journal of Psychology, 215*(3), 174–182. https://doi.org/10.1027/0044-3409.215.3.174
Lipsey, M., & Wilson, D. B. (2001). *Practical meta-analysis* (Vol. 49). CA: Thousand Oaks: SAGE Publications.
Lorch, R. F., Jr., Lorch, P. E., & Klusewitz, M. (1995). Effects of typographical cues on reading and recall of texts. *Contemporary Educational Psychology, 20*(1), 51–64. https://doi.org/10.1006/ceps.1995.1003
Mahdavi, A., & Azimi, S. (2012). The effects of cognitive strategies i.e. note-making and underlining on iranian efl learners' reading comprehension. *International Journal of Applied Linguistics and English Literature*, *1*(6), 1–6. https://doi.org/10.7575/ijalel.v.1n.6p.1
Mathews, C. O. (1938). Comparison of methods of study for immediate and delayed recall. *Journal of Educational Psychology, 29*(2), 101–106. https://doi.org/10.1037/h0055182
Mayer, R. E. (2021). Evidence-based principles for how to design effective instructional videos. *Journal of Applied Research in Memory and Cognition*. https://doi.org/10.1016/J.JARMAC.2021.03.007
Meyer, B. J. F., & Poon, L. W. (2001). Effects of structure strategy training and signaling on recall of text. *Journal of Educational Psychology, 93*(1), 141–159. https://doi.org/10.1037/0022-0663.93.1.141
Meyer, B. J. F., Brandt, D. M., & Bluth, G. J. (1980). Use of top-level structure in text: Key for reading comprehension of ninth-grade students. *Reading Research Quarterly, 16*(1), 72–103. https://doi.org/10.2307/747349
Miyatsu, T., Nguyen, K., & McDaniel, M. A. (2018). Five popular study strategies: Their pitfalls and optimal implementations. *Perspectives on Psychological Science, 13*(3), 390–407. https://doi.org/10.1177/1745691617710510
Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *BMJ, 339*, b2535. https://doi.org/10.1136/bmj.b2535
National Institute of Child Health and Human Development. (2000). *Report of the National Reading Panel. Teaching children to read: an evidence-based assessment of the scientific research literature on reading and its implications for reading instruction: Reports of the subgroups (NIH Publication No. 00–4754)*. Washington, DC.
Peterson, S. E. (1991). The cognitive functions of underlining as a study technique. *Reading Research and Instruction, 31*(2), 49–56. https://doi.org/10.1080/19388079209558078
Ponce, H. R., & Mayer, R. E. (2014a). An eye movement analysis of highlighting and graphic organizer study aids for learning from expository text. *Computers in Human Behavior, 41,* 21–32. https://doi.org/10.1016/j.chb.2014.09.010
Ponce, H. R., & Mayer, R. E. (2014b). Qualitatively different cognitive processing during online reading primed by different study activities. *Computers in Human Behavior, 30*, 121–130. https://doi.org/10.1016/j.chb.2013.07.054
Ponce, H., Mayer, R., Loyola, M. S., López, M. J., & Méndez, E. E. (2018a). When two computer-supported learning strategies are better than one: An eye-tracking study. *Computers and Education, 125*, 376–388. https://doi.org/10.1016/j.compedu.2018.06.024
Ponce, H. R., Mayer, R. E., Figueroa, V. A., & López, M. J. (2018b). Interactive highlighting for just-in-time formative assessment during whole-class instruction: effects on vocabulary learning and reading comprehension. *Interactive Learning Environments*, *26*(1) 42–60. https://doi.org/10.1080/10494820.2017.1282878
Rickards, J. P., & August, G. J. (1975). Generative underlining strategies in prose recall. *Journal of Educational Psychology, 67*(8), 860–865. https://doi.org/10.1037/0022-0663.67.6.860
Rickards, J. P., & Denner, P. R. (1979). Depressive effects of underlining and adjunt questions on children's recall of text. *Instructional Science, 8*, 81–90. https://doi.org/10.1007/BF00054983
Schneider, S., Beege, M., Nebel, S., & Rey, G. D. (2018). A meta-analysis of how signaling affects learning with media. *Educational Research Review, 23*, 1–24. https://doi.org/10.1016/j.edurev.2017.11.001

Schnell, T. R., & Rocchio, D. J. (1974). A study of the relative effectiveness of various underlining strategies on reading comprehension. In *Annual Meeting of the College Reading Association*. Retrieved from https://eric.ed.gov/?id=ED099821

Silvers, V. L., & Kreiner, D. S. (1997). The effects of pre-existing inappropriate highlighting on reading comprehension. *Reading Research and Instruction, 36*(3), 217–223. https://doi.org/10.1080/19388079709558240

Simard, D. (2009). Differential effects of textual enhancement formats on intake. *System, 37*(1), 124–135. https://doi.org/10.1016/j.system.2008.06.005

Stordahl, K. E., & Christensen, C. M. (1956). The effect of study techniques on comprehension and retention. *Journal of Educational Research, 49*(8), 561–570. https://doi.org/10.1080/00220671.1956.10882321

Suurmond, R., van Rhee, H., & Hak, T. (2017). Introduction, comparison, and validation of Meta-Essentials: A free and simple tool for meta-analysis. *Research Synthesis Methods, 8*(4), 537–553. https://doi.org/10.1002/jrsm.1260

Van Cog, T. (2022). The signaling (or cueing) principle in multimedia learning. In R. E. Mayer & L. Fiorella (Eds.), *The Cambridge handbook of multimedia learning* (3rd ed., pp. 221–230). Cambridge University Press.

Winchell, A., Lan, A., & Mozer, M. (2020). Highlights as an early predictor of student comprehension and interests. *Cognitive Science, 44*(11), e12901. https://doi.org/10.1111/cogs.12901

Wittrock, M. C. (1989). Generative processes of comprehension. *Educational Psychologist, 24*(4), 345–376. https://doi.org/10.1207/s15326985ep2404_2

Yue, C. L., Storm, B. C., Kornell, N., & Bjork, E. L. (2014). Highlighting and its relation to distributed study and students' metacognitive beliefs. *Educational Psychology Review, 27*(1), 69–78. https://doi.org/10.1007/s10648-014-9277-z

Authors and Affiliations

Héctor R. Ponce[1] · Richard E. Mayer[2] · Ester E. Méndez[3]

Richard E. Mayer
rich.mayer@psych.ucsb.edu

Ester E. Méndez
ester.mendez@usach.cl

[1] Faculty of Management and Economics, University of Santiago of Chile, Av. L. B. O'Higgins 3363, Santiago, Chile

[2] Department of Psychological and Brain Sciences, University of California, Santa Barbara, CA 93106-9660, USA

[3] Interactive Technology Lab, University of Santiago of Chile, Av. L. B. O'Higgins, 3363 Santiago, Chile